Resilient Emergency MANETs for Wilderness Safety

Directed Study Report

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Directed Study (CS 5976) — Advisors: Prof. Guevara and Prof. S. Basagni

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Code, data, and all cited artifacts:
`https://github.com/edoherty2017/resilient-emergency-manets-final`

Abstract

This directed study delivered a purpose-built, two-engine (Python/Rust) discrete-event simulation framework for year-scale energy and reliability analysis of a Meshtastic-style wilderness mesh — cross-validated on a shared micro-scenario between independently written engines (statewide parity pending), byte-reproducible, and released with hash-locked inputs — together with a Longley–Rice ITM terrain planning layer, a preregistration-grade field methodology, and a field acquisition system proven receiver-side end-to-end in the field. Its principal MODEL-ONLY insight: network survivability is governed by the idle-listening/duty-cycling policy, not the routing algorithm — five distinct always-on routing modes differ by $< 0.3\%$ in annual battery depletions while duty policies move depletions by $\approx$5.5–24$\times$, at a quantified cost in delivery ratio and SOS latency. The empirical program was adjusted in flight, with every adjustment registered before collection: Trial 1 served as a systems shakedown, and Trial 2 ran as a field campaign of four field days plus a fifth registered siting (2026-07-18 to 2026-07-26), re-planned after the beacon node's hardware failed, ending with an operationally successful receiver-only field day that produced two quantified findings substantiating the necessity of the controlled-beacon design (transmit-opportunity starvation; hop ambiguity). The planned calibration dataset was not obtained — 0 of the $\geq$2,500 planned points — and its acquisition is a small, fully specified next step: a $\approx$\$25 replacement beacon and one two-radio field day under the already-frozen protocol.

1 Original objectives and what changed

The original plan (preserved unedited in `docs/execution-roadmap-weekly.md`) had five success criteria:

Amendments and standing: the RSRP→ESP substitution is drafted but unsigned (disclosed, not assumed); the $\geq$40-opportunity rule is proposed, not approved; the study area is New Hampshire only; the planned Tuckerman/Huntington campaigns were replaced by the Ammonoosuc–Jewell system and then, in the field week, by the Moosilauke → Monadnock → Pack Monadnock sequence (§6, each step registered before collection). Radio operation across all field days used stock FCC-certified consumer hardware in its as-marketed configuration (§6.5); production-deployment authorization deferred (§7). The defensible scientific claim narrowed from “a validated emergency-network design” to “an uncalibrated planning and sensitivity framework plus a field-acquisition system awaiting controlled empirical validation.”

Original objective	Outcome
Architecture and design documentation	Delivered (`docs/system-architecture.md` and related).
≥2,500 empirical field points	**Not met — 0 calibration-grade points.** Trial 1 produced 0 calibration-eligible observations (§5.1); Trial 2's four field days (plus a fifth registered siting, §6) were ended by beacon hardware failure before any controlled link ran. A per-stratum replacement metric (≥40 scheduled opportunities per primary stratum) is a *proposed* amendment awaiting advisor review.
AIRMap calibration files	**Not produced** — blocked on controlled data; the gated pipeline that will produce them is built and tested.
10–15 page validation report (RMSE/MAE, heatmaps, failure matrix)	**Deferred, not abandoned.** All builder tooling exists; producible immediately after a two-radio field day (§8).
Large-scale simulation as optional appendix	**Inverted:** the simulation became the central completed contribution (§2–§5).

2 Simulator architecture

Two independent implementations of one model: `scripts/mesh_sim.py` (SimPy reference engine, detailed event traces) and `fastsim/` (Rust; summary-only engine for long horizons and multi-seed sweeps; nine shared modes — flood, min_hop, etx, energy_aware, lb_energy, duty_sync, duty_adaptive, rotate_lb, selective_duty; validated inputs; atomic output; deterministic tie-breaking). Randomness is keyed per phenomenon and entity, so workloads are comparable across algorithms and a congested MAC cannot alter the offered load.

Propagation/reception. Received power = 26.30 dBm reference (24.15 dBm TX EIRP + 2.15 dBi RX antenna gain + assumed 0 dB feed loss — the reference is *not* EIRP) minus a link-specific median loss, plus Gauss–Markov slow shadowing ($\sigma = 8$ dB, 30 s coherence) and a 2 dB per-packet fast term. Median loss: fixed↔fixed links use precomputed Longley–Rice ITM q50 over USGS 3DEP terrain; mobile↔fixed uses time-interpolated route-loss tables; mobile↔mobile uses an explicit heuristic (FSPL + 20 dB clutter + 12 dB/km beyond 1.5 km, 8 km cutoff) that is a workload model, not a fitted law. Reception requires RSSI ≥ -131 dBm (assumed, datasheet-derived, bench-unverified) and SNR ≥ -17.5 dB against a -114.0 dBm noise floor. All propagation constants are uncalibrated planning values.

MAC/airtime. LoRa SF11 / 250 kHz / CR 4/5 airtime; carrier-sense threshold -124 dBm; 40 ms slots; random DIFS/backoff; SOS priority; SNR-scaled flood contention; duplicate suppression; frozen-overlap collision resolution; 6 dB co-SF capture; half-duplex senders. Duty modes model per-second CAD preamble sampling requiring ≥2 preamble symbols of overlap. The engine separates *aggregate offered airtime* from *per-site physical occupancy* — conflating these invalidated all pre-audit "channel utilization" numbers.

Routing/duty policies. Routed modes build a multi-source shortest-path tree rooted at live gateways with per-mode edge costs. Duty assignments: uniform 5% (duty_sync); energy-runway-adaptive 2–25% (duty_adaptive); route-tree-awake (rotate_lb); backbone+articulation-awake (selective_duty). Routing never sees realized future weather.

Energy/solar/weather. 37 Wh batteries (85% usable); depletion/outage/revival state machine; 600 s packet TTL (engine constant, `fastsim/src/sim.rs`). Board currents (245 mA TX / 68 mA listen / 12 mA sleep / 25 mA GPS) are BENCH-CALIBRATE placeholders, not measurements. Solar: NOAA sun position, Haurwitz clear-sky, clearness-index split, 48-azimuth DEM horizon shading, 0.15 canopy transmittance below treeline, four-face 35° panels, 6 W relay panels at 75% system efficiency. One Mt. Washington-area ERA5-derived daily weather series applied statewide.

Mobility/logistics/traffic. Timestamped trail routes; kiosk rental model (20 bays, tiered demand, charge-gated checkout, nightly shuttle); traffic = telemetry + position beacons + hiker messaging + synthetic SOS incidents with logical duplicates and 5-minute fresh-packet retries to a gateway ACK or the retry ceiling. Traffic rates are scenario inputs, not measured demand.

3 Why a purpose-built simulator rather than ns-3

An ex-post engineering rationale (no contemporaneous ns-3 selection study exists in the repository), supporting only a narrow claim. The research variables are not primarily packet-forwarding mechanics: they are terrain-indexed loss tables, daily weather and solar horizon masks, battery death/revival, kiosk logistics, and route-popularity traffic — first-class objects in the custom model. The summary-only Rust engine runs year-scale multi-seed sweeps; keyed randomness gives paired workloads; the model natively reports the decision metrics (SOS tail latency, fleet availability, dark-site census, energy per delivered packet). Two independently written engines create a real opportunity to catch coding divergence (§4) — several semantic bugs were found this way.

The custom simulator has far less external validation and protocol depth than ns-3: it abstracts the PHY, carrier sensing, capture, CAD timing, buffering, retransmission, clock drift, and firmware queues, and shared model assumptions can survive in both engines. It does not prove compatibility with stock Meshtastic firmware. The defensible division of labor: purpose-built model for NH planning and sensitivity studies; cross-engine tests plus selected ns-3 or hardware-in-the-loop scenarios for MAC/protocol fidelity; a controlled field day plus bench measurements for empirical calibration.

4 Python/Rust cross-validation

Software regressions. `cargo test` at HEAD `2225c26` (2026-07-26): **47 unit + 3 integration tests, all passing**; log archived at `artifacts/sim/corrected/cargo_test_2026-07-26.log`.

Cross-process reproducibility. Four independent Rust-engine runs (min_hop, 1.3 days, seed 4242, hash-locked release_v1 inputs) produced byte-identical 419,738-byte outputs, SHA-256 `9d182b0d75cb…`; archived in `artifacts/sim/corrected/repro_check_2026-07-17/`.

Cross-engine agreement (micro-scenario). After reconciling seven semantic gaps in the Python engine, the shared pilot micro-scenario (3 days, seed 42, flood / duty_sync / selective_duty) passes **every pre-registered tolerance band on all six metrics in all three modes** (`artifacts/sim/corrected/micro_parity_2026-07-17.json`): maximum discrepancies 0.0045 absolute PDR (tolerance 0.03), 1.3% relative offered airtime (tolerance 10%), 0.3% relative duty-misses (tolerance 15%); deaths and SOS counts agree exactly. The release_v1 manifest's older parity remark predates this re-run and is superseded by it. An older nine-pair statewide comparison exists only in quarantine and must not be quoted as current validation.

Statewide two-engine parity: not yet run. The preregistered statewide parity (9 modes × seeds 42–46) remains pending; the micro-scenario comparison is the current substitute. In all cases,

engine agreement is a software check, not empirical validation — two implementations can agree on a wrong shared assumption.

5 Main findings

5.1 What Trial 1 established, and the model screens it motivated

1. **Trial 1 tested the collection system, not the propagation model.** The receiver log contains 686 records from 50 decoded node IDs in the afternoon hike window and 764 RF observations from 41 source IDs (24 with GPS) in the pre-hike soak (18 IDs in both windows; reconciled by `scripts/reconcile_trial1_counts.py`). **Zero** observations are calibration-eligible (`artifacts/airmap/live_trial/quality_gates.json`; its top-level `passed: true` reflects the relaxed operational ingest gate run with `require_calibration_grade=false` — the calibration gate itself records the failure, $0 < 30$ eligible rows): no hop-count telemetry, a co-located forwarder contaminating RSSI/geometry pairs, no controlled transmitter. This negative finding designed Trial 2's protocol (controlled beacon, sequence-number denominators, `hops_away == 0` eligibility) — a design whose necessity the Trial 2 campaign then substantiated empirically (§6.6).
2. **The free-space coverage story did not survive terrain modeling** (MODEL-ONLY; a model-to-model comparison, not a headline claim). Longley–Rice ITM over USGS 3DEP terrain reverses the FSPL screen on summit-facing links: Ammo relay→summit −132.2 dBm (below the assumed −131 dBm sensitivity; terrain-blocked path), Jewell relay→summit −118.7 dBm (marginal), while below-treeline relay→gateway links stay strong in the terrain-only model (−76.6 / −74.1 dBm; ITM excludes canopy loss, budgeted separately). Worst-case FSPL-vs-ITM disagreement: **62.3 dB** on the path-loss basis (ITM q50 158.5 dB vs in-artifact FSPL 96.2 dB). Collector-gap coverage at ITM q90: 100% of 84 sampled points above raw sensitivity but only **45.2%** above the −100 dBm planning threshold — "the gap would have been covered" is downgraded to *marginal, unproven.*
3. **The statewide planning topology fails its controlling screen** (MODEL-ONLY; internal model-to-model check). With 52 uncalibrated short-link substitutions excluded, the −100 dBm planning screen reports 87 of 217 sites stranded and 15 of 25 routes below 85% modeled coverage; even restoring the 52 links leaves 53 sites stranded.

5.2 Corrected year-scale simulation results (model-only, single-engine)

Nine modes × seeds 42–46 × 365 days on the corrected Rust engine (`artifacts/sim/corrected/release_v1/`, hash-locked inputs; values are means across five seeds, so SOS counts are fractional):

All PDR 95% CIs are below ±0.003 and deaths-per-year CIs are similarly tight: given a seed the engine is near-deterministic, so the dominant uncertainty is the uncalibrated model, not seed noise. Raw counts hid two things: (a) always-on modes leave ∼46% of solar fleet-time dark — 118 of 159 solar sites are available <90% of the year; (b) uniform duty_sync's 94.7% SOS delivery conceals a 35-minute p95 latency, while backbone-awake duty modes hold 33–62 s p95 at 99.3–99.5% delivery. Every mode retains rare 1–2 h worst-case SOS incidents (retry-ceiling limit). **No overall winner is declared**: delivery, availability, and SOS tail latency still trade off, statewide two-engine parity is pending, and the model is uncalibrated.

Mode	PDR	Deaths/yr	SOS del.	SOS p95	Fleet avail.	Sites <90%	mWh/pkt
flood	0.911	29,234	189.2/189.2 (100%)	32 s	53.5%	118	1.44
min_hop	0.862	29,164	189.2/189.2 (100%)	5 s	53.7%	118	0.42
etx	0.866	29,165	189.2/189.2 (100%)	31 s	53.7%	118	0.42
energy_aware	0.866	29,164	189.2/189.2 (100%)	4 s	53.7%	118	0.42
lb_energy	0.866	29,164	189.0/189.2 (99.9%)	5 s	53.7%	118	0.42
duty_sync	0.728	**1,217**	179.2/189.2 (94.7%)	**35.0 min**	**96.5%**	**7**	0.30
duty_adaptive	0.755	1,737	187.8/189.2 (99.3%)	15.0 min	94.7%	13	0.40
rotate_lb	0.809	4,877	188.2/189.2 (99.5%)	**33 s**	91.0%	31	0.35
selective_duty	0.811	5,311	187.8/189.2 (99.3%)	**62 s**	89.7%	33	0.36

Table 1: MODEL-ONLY year-scale results (`release_v1/corrected_stats.json`, `release_v1/meaningful_metrics.json`).

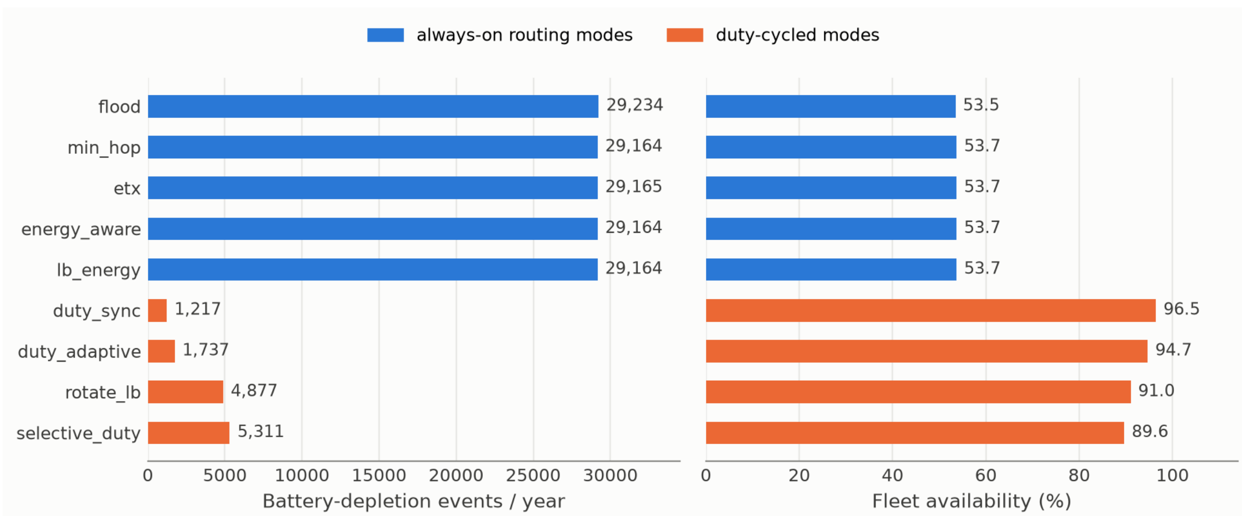

Figure 1: Year-scale survivability by mode (MODEL-ONLY, uncalibrated; 9 modes × seeds 42–46, 365 days). Left: annual battery-depletion events — the five always-on routing modes are indistinguishable (< 0.3% spread) while duty policies cut depletions ≈5.5–24×. Right: fleet availability.

5.3 Central insight: duty policy dominates survivability, not routing

The five always-on modes span materially different routing strategies, yet their annual depletion events sit within a **0.3% band** (29,164–29,234/yr), and four of five share identical availability (53.7%) and dark-site counts (118). Switching the *duty/wake policy* while holding the routing family fixed moves deaths by **≈ 5.5–24×** (to 1,217–5,311/yr) and fleet availability from ∼54% to 90–96%. The energy budget is dominated by idle listening (68 mA modeled listen vs 12 mA sleep); what a node does while *not* forwarding matters far more to survival than which tree its packets follow. If this survives calibration, solar relay hardware and firmware should be selected around the wake/CAD architecture, with routing sophistication a second-order optimization. The quantified cost — PDR −0.05 to −0.14 vs the routed always-on modes (−0.10 to −0.18 vs flood) plus the SOS-latency structure above — makes this an explicit, tunable trade, not a free win. The separation is bound to release_v1 pending regeneration (§7); the concrete reason it is unlikely to be an engine artifact is scale — the 5.5–24× effect dwarfs the worst cross-engine discrepancy observed on the repaired engine (≤ 1.3%, §4). (MODEL-ONLY; the listen/sleep currents are bench-calibration placeholders — precisely why the bench test and a controlled field day are the critical path.)

6 Trial 2: the field campaign

The causal frame, stated plainly: the two-radio controlled-beacon protocol could not be executed because the beacon node's hardware failed during the field week — its internal battery would not hold charge, and its USB-C charging port then broke. Trial 2 was therefore adjusted — with every adjustment registered before the data it affected was collected — to a receiver-only field day scoring prospective public-station contact predictions. The controlled-beacon prediction packs remain frozen and unscored for a future two-radio day.

6.1 Preregistration and freezes

The strongest freeze in the record is the 2026-07-19 preregistration (`artifacts/trial2/prereg_manifest.json`): 12 hashed inputs — prediction tables, generator, DEM manifest, radio metadata, eligibility gate, analysis code — bound at git `3d15a57` with a clean worktree, before any calibration-eligible data existed. (Disclosure: 22 rig-bringup telemetry rows tagged `trial2-moosilauke-20260719`, containing no beacon packets, predate the freeze timestamp by minutes.) The field-day prediction pack for Moosilauke/Kearsarge/Monadnock is registered via SHA-256 digests recorded in dated siting documents (2026-07-23; CSV `a85e4d08…`, manifest `3e6df03f…`), re-verified matching on 2026-07-26; these registrations were git-timestamped by commit `1229309` on 2026-07-26.

6.2 Campaign log: attempts 1–3 (Jul 18–21)

Sat Jul 18: the rig never ran (0 GPS, 0 telemetry rows). Sun Jul 19 (Moosilauke): preserved GPS and telemetry evidence covers 09:10–10:19 EDT (≈1.1 h; 35,891 NMEA sentences), with the receiver radio's serial live only 09:14–10:13 EDT; **zero beacon packets — the beacon was already broken**. Tue Jul 21 (retry): GPS logged throughout the ≈3-hour attempt (13:21–16:23 UTC, 110,115 sentences); the radio never enumerated on USB (power-only cable; 0 radio rows). All raw evidence is preserved in `artifacts/trial2/raw_pull_20260721/` (`telemetry_stream.jsonl`: 65,271 lines, 65,268 parseable JSON rows; `nmea_stream.jsonl`: 190,680 lines; SHA256SUMS added 2026-07-26). Per the frozen protocol's own definition these are operational failures with preserved evidence.

6.3 The siting-decision basis and the re-siting criterion

With the beacon dead, the only possible RF sources were public Meshtastic stations, so the field day had to sit inside the live public-mesh region. A checksummed snapshot of the regional community's live dashboard (2026-07-23, `artifacts/trial2/nhmesh_live_snapshot_20260723/`) showed the ecosystem had largely migrated to a different protocol: **544 of 550 recently-positioned nodes were MeshCore — a protocol Meshtastic radios cannot receive — leaving 6 Meshtastic stations on the map, none in the White Mountains** (point-in-time caveat, stated plainly: this was a collector-view snapshot, not an ecosystem census — a same-dashboard pull on 2026-07-26 12:20 UTC, archived at `artifacts/trial2/nhmesh_activity_20260726/`, shows 276 positioned Meshtastic nodes seen within 12 h; the 2026-07-23 snapshot is preserved because it is what the siting decision was made on). ITM feasibility ranking over the statewide DEM selected Mt. Monadnock, then — when a site policy constraint ruled Monadnock State Park out — Pack Monadnock (Miller State Park). Both sitings were registered in dated documents before their field days.

6.4 Monadnock: registered, never executed

The Monadnock pack (beacon point 42.86010/ − 72.10682 at 909 m by the ≥850 m first-ascent rule; 33 route samples; 8 prediction rows appended with prior sites verified byte-identical; DEM sha `57936e6b…`; contact predictions `monadnock_livemesh_predictions_20260723.json`, sha `aba23c13…`) was fully registered on 2026-07-23 and never walked. It stands as a frozen sibling registration.

6.5 Pack Monadnock (2026-07-26): the first operationally successful day

Executed under `trial2-packmonadnock-20260726` on the OSM-routed Wapack loop. The rig ran end-to-end: radio logging all day, 1 Hz GPS, summit reached (714.6 m raw GGA maximum, 13:50:13 UTC) with a 13:31–14:04 UTC dwell within 60 vertical meters of the maximum. Raw evidence: `artifacts/trial2/raw_pull_20260726/` with SHA256SUMS. Trial-ID bookkeeping is disclosed in full: the rig was powered at home from 10:01 UTC; 8 field-day boot rows (10:01–10:06 UTC) plus 55 rows from a 2026-07-22 bench session carry the superseded `trial2-moosilauke-20260722` tag (the field tag activates at 10:07:38 UTC); 27 pre-departure rows and 8 post-hike drive-tail rows carry the field tag.

Radio operation, all field days: stock FCC-certified consumer hardware (Heltec V3 receiver), unmodified firmware and transmit power, standard US 915 MHz ISM-band frequencies (LongFast defaults), default channel-access behavior, the default public channel carrying the project's own traffic, all equipment placed and retrieved the same day.

Registered contact predictions and outcomes — scored in two provenance tiers, never aggregated (the receiver's decode capability is independently evidenced: it logged foreign frames during the campaign, including one during the summit dwell at −117 dBm):

Tier 1 — pre-departure registered pack (`packmonadnock_livemesh_predictions_20260726.json`, sha `71db9f00…e9e2`; recorded pre-collection in the siting document in truncated form, full digest in its addendum): Greenville, CONTACT_PREDICTED at −87.9 dBm best-along-route — **not heard (0/1)**. Keene Court St, NO_CONTACT_PREDICTED at −144.4 dBm — **null held (1/1)**. Three MARGINAL links — none heard.

Tier 2 — trailhead supplement (`packmonadnock_livemesh_supplement_20260726.json`, sha `f50bda83…53c5f`; provenance explicitly weaker: digest first durably recorded post-collection, pre-walk timestamp self-asserted): three additional CONTACT_PREDICTED links (−90.2 to −91.1 dBm) — none heard; four additional predicted nulls — all held.

Null confirmations are scored but carry little evidential weight: the same transmit censoring that voids the contact misses (§6.6) applies symmetrically — a station that does not transmit produces a held null regardless of link physics — and the Keene prediction (−144.4 dBm) lies ≈13 dB below the assumed decode floor, making its null near-trivial.

6.6 Field findings: the empirical case for the controlled beacon

The contact misses are **transmit-censored, and the campaign measured the censoring**:

1. **Transmit-opportunity starvation.** The public stations' measured transmit cadence (60-minute activity window generated 2026-07-26 15:02:52 UTC; archived with checksums at `artifacts/trial2/nhmesh_activity_20260726/`; an adjacent-hour estimate, ≈14:03–15:03

UTC, assumed representative of the dwell window) is 0–4 packets per hour per station — Greenville sent 1 packet in the archived hour, i.e. ≈0.6 expected transmissions during the 33-minute GPS-verified summit dwell (13:31–14:04 UTC); two of the three Tier-2 CONTACT-predicted stations sent 0. (A scoreable opportunity = a station transmission while the receiver sits inside that link's predicted-contact route segment.) The receiving side was equally starved in reverse: the project node kept stock broadcast cadence and was never heard by any community collector all day. A link prediction cannot be confirmed by a station that does not transmit during the window — at these cadences the expected number of scoreable opportunities per field day is approximately zero, regardless of link quality.

2. **Hop ambiguity.** All six origin-decoded foreign frames logged across the day carry `hops_away` between 1 and 5 (or undecodable hop fields) — **zero direct receptions among origin-decoded frames** (one additional frame with undecodable origin carries `hops_away=0`; disclosed, not scored). Mesh-relayed packets attribute their RSSI to an unknown last-hop transmitter, not to the origin path, so none of the six decodes is usable as a path-loss observation. (Systems observation, reported as such: packets originating at stations 86–88 km away reached the project radio through the mesh — both such decodes logged in the post-hike vehicle segment — multi-hop delivery functioning in the wild.)

Together these two measurements are the empirical justification for the frozen protocol's two central rules — a controlled beacon at 30 s cadence (defeats opportunity starvation) and `hops_away == 0` eligibility (defeats hop ambiguity). Opportunistic validation against a public mesh fails not on RF physics but on opportunity statistics and attribution; the controlled two-radio design is necessary, not merely preferable.

6.7 The remaining step

Zero calibration-eligible strata exist; the ≥40-opportunity target was never reached on any day; the Moosilauke, Kearsarge, and Monadnock beacon packs remain frozen and unscored. None of this is dropped or relabeled. The concrete remaining step is small and fully specified: a replacement beacon board (≈$25), the already-frozen protocol, and one two-radio field day (§8). The raw-dataset package item was built 2026-07-26: `artifacts/dataset_release/evidence-2026-07-26/` — 128,721 observations, 0 calibration-grade, with `MANIFEST.json` and data dictionary.

7 Limitations

Empirical (the binding constraint). No controlled dataset exists; Trial 1 and the Trial 2 campaign together yield zero calibration-grade observations, so the project has no held-out field estimate of RSSI error, PDR error, or shadow variance. Receiver sensitivity (−131 dBm), feed/body/foliage losses, board currents, cold-battery capacity, and panel yield are assumptions or placeholders. Public-mesh observations are non-calibration-grade by construction (unknown station EIRP/antennas/uptime) and are never used as path-loss data. A future controlled field day calibrates propagation and direct-link PDR on its sampled routes; it cannot by itself validate statewide traffic, routing, kiosk, or annual energy behavior. Field sites are restricted: one shakedown (Mt. Washington) and one operationally successful receiver-only day (Pack Monadnock), summer only.

Modeling. ITM q50 medians without a validated residual model; q90 route arrays absent. $\sigma =$ 8 dB shadowing, 2 dB fast fade, capture threshold, mobile–mobile heuristic, synthetic traffic/SOS rates, and the 600 s TTL are assumptions. The MAC is not stock Meshtastic firmware. One statewide weather series; simplified solar geometry; geology priors in the repository are uncited placeholders not wired into any prediction and are not model inputs here. The purpose-built simulator is not independently validated through a widely used third-party framework (ns-3 spot-validation is future work).

Software, release, and provenance. Statewide two-engine parity is pending; corrected-labeled artifacts outside `release_v1/` are stale and marked accordingly. release_v1 was generated 2026-07-14 from a dirty worktree (its manifest records this) and carries a disclosed manifest inconsistency (git `d37172c` vs `f5e83f0`; the four locked input hashes agree in both); regeneration (release_v2) is incomplete (5 of 9 modes, no manifest) and **not citable** — the §5.2 numbers are bound to the archived release_v1 artifacts. Post-2026-07-19 Trial 2 registrations were committed 2026-07-26 (`1229309`); their pre-commit provenance was in-doc SHA-256 digests, as disclosed in the claim-policy note. Authorization for a production fixed-relay deployment remains an open question (`docs/fcc-part15-compliance-memo.md`); field-day operation used certified consumer hardware as marketed (§6.5). Passing tests bound, but do not prove, the absence of defects.

8 Future work

(1) **The two-radio field day** (the critical path): replacement beacon board (≈\$25), the already-frozen protocol (30 s cadence, sequence numbers, `hops_away == 0`, ≥40 opportunities per primary stratum), scoring the frozen Moosilauke and/or Monadnock packs; the output feeds the original validation deliverable (RMSE/MAE, predictive-vs-actual heatmaps, failure matrix) — all tooling exists. (2) Bench calibration of the BENCH-CALIBRATE constants. (3) Complete release_v2 with a full manifest and regenerate §5.2 from it. (4) Statewide two-engine parity per the preregistered tolerance plan. (5) ns-3 spot-validation of selected MAC/protocol scenarios. (6) Advisor-decided amendments: the RSRP→ESP substitution and the ≥40-opportunity rule. (7) Dataset-release packaging: completed 2026-07-26 (`artifacts/dataset_release/evidence-2026-07-26/`).

9 Data and reproducibility statement

All quantitative statements in this report are artifact-backed measurements, archived software checks, or simulation results labeled MODEL-ONLY. Simulation numbers cite the corrected multi-seed release (`release_v1`, hash-locked inputs); superseded development-phase outputs are archived separately in the repository (`WITHDRAWN-DO-NOT-CITE/`, with a manifest) and are not cited here, per the acceptance requirement that superseded numbers be clearly archived and not mixed with final results. Same-seed engine output is byte-identical (§4). Trial 2 registrations made after 2026-07-19 are git-timestamped by commit `1229309`; every SHA-256 digest cited in this report was re-verified against its artifact on 2026-07-26. All artifacts cited by path in this report resolve in the project repository, `https://github.com/edoherty2017/resilient-emergency-manets-final` (branch `main`, release `directed-study-final`).

A Frozen prediction packs and scoring ledger

Propagation method detail for all packs: Longley–Rice ITM q50 point-to-point over USGS 3DEP DEMs, 26.30 dBm receiver-power reference (the registration files label this quantity “EIRP ref” — a terminology carry-over in those frozen documents, not a numerical discrepancy; the value is

Pack	Registered	Status
Mt. Washington (Ammo/Jewell)	2026-07-19 freeze, git `3d15a57`, clean worktree	Deferred — never walked; retained
Moosilauke (Gorge Brook; beacon 44.01708/−71.83777 @1352 m)	fieldday pack, digests recorded 2026-07-23	Frozen, unscored (beacon failure); awaits two-radio day
Kearsarge (Winslow; beacon 43.38735/−71.85970 @771 m)	same pack	Deferred (ITM predicts no public-mesh contact from site); retained
Monadnock (White Dot; beacon 42.86010/−72.10682 @909 m)	pack + livemesh JSON (`aba23c13…`), 2026-07-23	Registered, never executed (site policy); retained
Pack Monadnock (Wapack loop)	livemesh JSON (`71db9f00…`) pre-departure; supplement (`f50bda83…`) weaker tier	**Executed 2026-07-26.** Tier 1: 0/1 contact, 1/1 null held. Tier 2: 0/3 contacts, 4/4 nulls held. Misses transmit-censored (§6.6); evidence basis: complete Pi instrument log

identical), receiver height 1.5 m, station heights as documented per pack (assumed stock where unpublished, labeled in the registration files).

B Design implications: sensitivity of the survivability result

This appendix responds to the advisor's request (2026-07-30) to translate the idle-listening finding of §5.3 into engineering terms. All values are MODEL-ONLY (uncalibrated constants, §7); the new sweeps ran on the current repaired engine against the release_v1 frozen inputs (17 runs, 365 days each; hash manifest at `artifacts/sim/appendixB_sweeps/`). The 68 mA baseline reproduces the release_v1 depletion count within 0.04% (29,175 vs. 29,164 at seed 42), while its PDR differs modestly (0.843 vs. 0.866) reflecting post-release engine repairs — comparisons below are therefore made within the sweep.

B.1 — What listen current makes always-on viable? Sweeping the router listen current under always-on energy-aware routing (stock 37 Wh battery, 6 W panel):

Listen current	Depletions/yr	Fleet availability	PDR
68 mA (ESP32-class, stock)	29,175	53.5%	0.843
40 mA	13,647	72.4%	0.836
20 mA	2,824	90.6%	0.832
10 mA	**27**	**99.8%**	0.831
5 mA	0	100.0%	0.831
2 mA (nRF52-class)	0 (3 seeds)	100.0%	0.832

Table 2: Listen-current sweep (MODEL-ONLY; energy_aware, seed 42; the 5 and 2 mA rows verified at seeds 43–44).

Always-on operation becomes viable at ∼10 mA on the stock energy kit, and at the nRF52-class 2–5 mA receive currents the modeled fleet loses no node all year — while delivery is essentially flat across the sweep. In this model, sufficiently low listen current dominates every duty-cycling policy on both axes at once, which quantifies the hardware target: bring the radio's idle receive path under ~10 mA and the duty-versus-delivery trade of §5.3 dissolves. (It also identifies listen current as the first constant to pin at the bench, and makes the wake-up-receiver direction a natural follow-on study.)

B.2 — What capacity reaches 90–95% availability at stock current? Holding 68 mA and sweeping the energy kit:

Battery	Panel	Depletions/yr	Fleet availability	PDR
37 Wh (stock)	6 W (stock)	29,175	53.5%	0.843
74 Wh	6 W	14,856	54.9%	0.842
148 Wh	6 W	6,546	57.2%	0.842
74 Wh	12 W	9,657	77.9%	0.835
148 Wh	12 W	4,014	79.9%	0.835
296 Wh	12 W	1,776	84.0%	0.834
148 Wh	24 W	1,551	**94.6%**	0.832
296 Wh	24 W	235	**98.5%**	0.831

Table 3: Battery × panel sweep at stock 68 mA listen current (MODEL-ONLY; energy_aware, seed 42).

Availability is panel-dominated: at the stock 6 W panel, even 4× battery moves availability only 53.5%→57.2%, because winter harvest — not storage — is binding. Reaching the 90–95% band at stock listen current requires roughly 4× battery *and* 4× panel (148 Wh / 24 W → 94.6%); 8×/4× reaches 98.5%. Per relay, that capacity route is materially more expensive than the listen-current route of B.1, which reaches higher availability on the stock kit — the design lever is the receiver, not the battery.

B.3 — What duty cycle preserves acceptable SOS latency? From the released policy comparison (§5.2, no new runs): the threshold is structural rather than a percentage. Uniform low-duty (duty_sync, 5%) buys the largest survivability gain (24×) but crosses the latency cliff — 94.7% SOS delivery with a 35-minute p95, because deliveries ride the 5-minute retry loop. Backbone-awake policies (rotate_lb, selective_duty), which keep the current routing tree's relays listening and duty-cycle everyone else, hold SOS p95 at 33–62 s with 99.3–99.5% delivery while retaining a 5.5–6× survivability gain. Within this model, any deployment with an SOS-latency requirement should duty-cycle *off-tree* nodes only; uniform duty cycling is a survivability tool for networks without latency-critical traffic.